

$$\Phi(p) = 2p' + x^2 p''$$

Uvažujme bázi V $P = (1, x, x^2)$ a vypočítajme $\Phi(1), \Phi(x), \Phi(x^2)$:

- $\Phi(1) = 2 \cdot 0 + x^2 \cdot 0 = 0$
- $\Phi(x) = 2 \cdot 1 + x^2 \cdot 0 = 2$
- $\Phi(x^2) = 2 \cdot 2x + x^2 \cdot 2 = 4x + 2x^2$

Matice operátora vzhledem $\mathcal{Q}$ $\mathcal{B}$ je:

$$[\Phi]_{\mathcal{B}}^{\mathcal{P}} = \begin{bmatrix} 0 & 2 & 0 \\ 0 & 0 & 4 \\ 0 & 0 & 2 \end{bmatrix}$$

Najdeme vl. čísla operátora Φ

$$\det([\Phi]_{\mathcal{B}}^{\mathcal{B}} - \lambda I_3) = \begin{vmatrix} -\lambda & 2 & 0 \\ 0 & -\lambda & 4 \\ 0 & 0 & 2-\lambda \end{vmatrix} = (-\lambda)^2(2-\lambda)$$

$$\begin{aligned} \lambda_1 &= 2 \\ \lambda_2 &= 0 \end{aligned}$$

$$M_{\lambda_1} : \left[\begin{array}{ccc|c} -2 & 2 & 0 & 0 \\ 0 & -2 & 4 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right] \quad \begin{aligned} -2x_2 + 4x_3 &= 0 \\ x_2 &= 2x_3 \end{aligned} \quad \begin{aligned} -2x_1 + 2x_2 &= 0 \\ x_1 &= x_2 \end{aligned} \quad \left. \vphantom{\begin{array}{ccc|c} -2 & 2 & 0 & 0 \\ 0 & -2 & 4 & 0 \\ 0 & 0 & 0 & 0 \end{array}} \right\} M_{\lambda_1} = \text{Span} \left\{ \begin{pmatrix} 2 \\ 2 \\ 1 \end{pmatrix} \right\}$$

$$M_{\lambda_2} : \left[\begin{array}{ccc|c} 0 & 2 & 0 & 0 \\ 0 & 0 & 4 & 0 \\ 0 & 0 & 2 & 0 \end{array} \right] \quad \begin{aligned} x_2 &= 0 \\ x_3 &= 0 \\ x_1 &= t \end{aligned} \quad \left. \vphantom{\begin{array}{ccc|c} 0 & 2 & 0 & 0 \\ 0 & 0 & 4 & 0 \\ 0 & 0 & 2 & 0 \end{array}} \right\} M_{\lambda_2} = \text{Span} \left\{ \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \right\}$$

Jordanove reťazky budú tvoriť:

$$\begin{array}{ccc} \vec{v}_1^1 & \xrightarrow{\phi - 2\text{id}_V} & \vec{0} \\ \vec{v}_2^2 \xrightarrow{\phi - \text{id}_V} \vec{v}_1^2 & \xrightarrow{\phi - 0\text{id}_V} & \vec{0} \end{array}$$

za vektor $\vec{v}_1^1$ zvolíme ľubovoľný nenulový vektor z M_{λ_1} napr. $\begin{pmatrix} 2 \\ 1 \end{pmatrix}$ a za vektor $\vec{v}_1^2$ zvolíme ľubovoľný nenulový vektor z M_{λ_2} napr. $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$

Keďže $\vec{v}_1^2 \in \text{Im}(\phi - 0\text{id}) = \text{Im} \phi$ $\vec{v}_2^2$ vieme nájsť riešením:

$$\left[\begin{array}{ccc|c} 0 & 2 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 2 & 0 \end{array} \right] \sim \left[\begin{array}{ccc|c} 0 & 1 & 0 & 1/2 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right] \text{ napr. vektor } \vec{v}_2^2 = \begin{pmatrix} 0 \\ 1/2 \\ 0 \end{pmatrix}$$

Prevedieme $\vec{v}_1^1, \vec{v}_1^2, \vec{v}_2^2$ späť na polynómy:

$$\left. \begin{array}{l} \vec{v}_1^1 = 2 + 2x + x^2 \\ \vec{v}_1^2 = 1 \\ \vec{v}_2^2 = \frac{1}{2}x \end{array} \right\} B = (1, \frac{1}{2}x, 2 + 2x + x^2) \text{ tvoří bázu } V$$

Matrica operátora vzhľadom k B je:

$$[\phi]_B^B = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 2 \end{bmatrix}$$

$$\begin{aligned} \text{alebo ... } \phi(\vec{v}_1^2) &= 0 \vec{v}_1^2 = \vec{0} \\ \phi(\vec{v}_2^2) &= 0 \vec{v}_2^2 + \vec{v}_1^2 = \vec{v}_1^2 \\ \phi(\vec{v}_1^1) &= 2 \vec{v}_1^1 \end{aligned}$$